

Mongo DB

Install Mongo DB → <https://www.mongodb.com/docs/manual/installation/>

[Mongodb.com/download/community](https://www.mongodb.com/download/community)

```
Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\Shaun> mongosh
Current Mongosh Log ID: 622a370cf0f5c85ca2b78a20
Connecting to:      mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+1.2.3
Using MongoDB:      5.0.6
Using Mongosh:      1.2.3

For mongosh info see: https://docs.mongodb.com/mongodb-shell/

-----
The server generated these startup warnings when booting:
  2022-03-10T17:27:37.052+00:00: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----
```

```
Kyle Cook
$ mongosh
Current Mongosh Log ID: 61127e47c24b4afb0c4ccae2
Connecting to:      mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
Using MongoDB:      5.0.2
Using Mongosh:      1.0.4

For mongosh info see: https://docs.mongodb.com/mongodb-shell/

-----
The server generated these startup warnings when booting:
  2021-08-07T09:20:13.164-05:00: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

test> show dbs
admin      41 kB
appdb      49.2 kB
config     73.7 kB
local     73.7 kB
test> use appdb
switched to db appdb
appdb> show collections
users
appdb> db.dropDatabase()
{ ok: 1, dropped: 'appdb' }
appdb> show dbs
```

```

test> use appdb
switched to db appdb
appdb> show dbs
admin    41 kB
config  73.7 kB
local   73.7 kB
appdb> db
appdb
appdb> db.users.insertOne({ name: "John" })
{
  acknowledged: true,
  insertedId: ObjectId("61127fc57ec144c25156c4d9")
}
appdb> db.users.find()
[ { _id: ObjectId("61127fc57ec144c25156c4d9"), name: 'John' } ]
appdb> db.users.insertOne({ name: "Sally", age: 19, address: { street: "987 North St" }, hobbies: ["Running"] })
{
  acknowledged: true,
  insertedId: ObjectId("611280447ec144c25156c4da")
}
appdb>

```

```

{
  acknowledged: true,
  insertedId: ObjectId('6a0848d2055299d2b6abc114')
}

```

```

{
  acknowledged: true,
  insertedId: ObjectId('6a084984055299d2b6abc115')
}

```

```

appdb> db.users.find()
[
  { _id: ObjectId("61127fc57ec144c25156c4d9"), name: 'John' },
  {
    _id: ObjectId("611280447ec144c25156c4da"),
    name: 'Sally',
    age: 19,
    address: { street: '987 North St' },
    hobbies: [ 'Running' ]
  }
]

```

```

appdb> db.users.find ()
[
  { _id: ObjectId('6a0848d2055299d2b6abc114'), name: 'John' },
  {
    _id: ObjectId('6a084984055299d2b6abc115'),
    name: 'Sally',
    age: 19,
    address: { street: ' 1 Adam Road' },
    hobbies: [ 'Running' ]
  }
]

```

```
appdb> db.users.insertMany([ { name: "Jill" }, { name: "Mike" } ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId("611280837ec144c25156c4db"),
    '1': ObjectId("611280837ec144c25156c4dc")
  }
}
appdb> db.users.find()
[
  { _id: ObjectId("61127fc57ec144c25156c4d9"), name: 'John' },
  {
    _id: ObjectId("611280447ec144c25156c4da"),
    name: 'Sally',
    age: 19,
    address: { street: '987 North St' },
    hobbies: [ 'Running' ]
  },
  { _id: ObjectId("611280837ec144c25156c4db"), name: 'Jill' },
  { _id: ObjectId("611280837ec144c25156c4dc"), name: 'Mike' }
]
```

```
appdb> db.users.insertMany([ { name: "Kyle", age: 26, hobbies: [ "Weight Lifting", "Bowling" ], address: { street: "123 Main St", city: "New York City" } }, { name: "Billy", age: 41, hobbies: [ "Swimming", "Bowling" ], address: { street: "442 South St", city: "New York City" } } ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId("6112809d7ec144c25156c4dd"),
    '1': ObjectId("6112809d7ec144c25156c4de")
  }
}
```

```
appdb> db.users.find()
[
  { _id: ObjectId("61127fc57ec144c25156c4d9"), name: 'John' },
  {
    _id: ObjectId("611280447ec144c25156c4da"),
    name: 'Sally',
    age: 19,
    address: { street: '987 North St' },
    hobbies: [ 'Running' ]
  },
  { _id: ObjectId("611280837ec144c25156c4db"), name: 'Jill' },
  { _id: ObjectId("611280837ec144c25156c4dc"), name: 'Mike' },
  {
    _id: ObjectId("6112809d7ec144c25156c4dd"),
    name: 'Kyle',
    age: 26,
    hobbies: [ 'Weight Lifting', 'Bowling' ],
    address: { street: '123 Main St', city: 'New York City' }
  },
  {
    _id: ObjectId("6112809d7ec144c25156c4de"),
    name: 'Billy',
    age: 41,
    hobbies: [ 'Swimming', 'Bowling' ],
    address: { street: '442 South St', city: 'New York City' }
  }
]
```


Find the 1st and 2nd collections as follow:

```
appdb> db.users.find().limit(2)
[
  { _id: ObjectId("61127fc57ec144c25156c4d9"), name: 'John' },
  {
    _id: ObjectId("611280447ec144c25156c4da"),
    name: 'Sally',
    age: 19,
    address: { street: '987 North St' },
    hobbies: [ 'Running' ]
  }
]
appdb> █
```

```
appdb> db.users.find().sort({ name: 1 }).limit(2)
[
  {
    _id: ObjectId("6112809d7ec144c25156c4de"),
    name: 'Billy',
    age: 41,
    hobbies: [ 'Swimming', 'Bowling' ],
    address: { street: '442 South St', city: 'New York City' }
  },
  { _id: ObjectId("611280837ec144c25156c4db"), name: 'Jill' }
]
appdb> db.users.find().sort({ name: 1 }).limit(2)
```

```
appdb> db.users.find().sort({ name: -1 }).limit(2)
[
  {
    _id: ObjectId("611280447ec144c25156c4da"),
    name: 'Sally',
    age: 19,
    address: { street: '987 North St' },
    hobbies: [ 'Running' ]
  },
  { _id: ObjectId("611280837ec144c25156c4dc"), name: 'Mike' }
]
```

```
appdb> db.users.find().sort({ age: 1, name: -1 }).limit(2)
[
  { _id: ObjectId("611280837ec144c25156c4dc"), name: 'Mike' },
  { _id: ObjectId("61127fc57ec144c25156c4d9"), name: 'John' }
]
```

```
appdb> db.users.find().sort({ age: -1, name: -1 }).limit(2)
[
  {
    _id: ObjectId("6112809d7ec144c25156c4de"),
    name: 'Billy',
    age: 41,
    hobbies: [ 'Swimming', 'Bowling' ],
    address: { street: '442 South St', city: 'New York City' }
  },
  {
    _id: ObjectId("6112809d7ec144c25156c4dd"),
    name: 'Kyle',
    age: 26,
    hobbies: [ 'Weight Lifting', 'Bowling' ],
    address: { street: '123 Main St', city: 'New York City' }
  }
]
```

```
appdb> db.users.find().limit(2)
[
  { _id: ObjectId("61127fc57ec144c25156c4d9"), name: 'John' },
  {
    _id: ObjectId("611280447ec144c25156c4da"),
    name: 'Sally',
    age: 19,
    address: { street: '987 North St' },
    hobbies: [ 'Running' ]
  }
]
appdb> db.users.find().skip(1).limit(2)
[
  {
    _id: ObjectId("611280447ec144c25156c4da"),
    name: 'Sally',
    age: 19,
    address: { street: '987 North St' },
    hobbies: [ 'Running' ]
  },
  { _id: ObjectId("611280837ec144c25156c4db"), name: 'Jill' }
]
appdb> █
```

```
appdb> db.users.find({ name: "Kyle" })
[
  {
    _id: ObjectId("6112809d7ec144c25156c4dd"),
    name: 'Kyle',
    age: 26,
    hobbies: [ 'Weight Lifting', 'Bowling' ],
    address: { street: '123 Main St', city: 'New York City' }
  }
]
appdb> █
```

```

appdb> db.users.find({ name: "Kyle" })
[
  {
    _id: ObjectId("6112809d7ec144c25156c4dd"),
    name: 'Kyle',
    age: 26,
    hobbies: [ 'Weight Lifting', 'Bowling' ],
    address: { street: '123 Main St', city: 'New York City' }
  }
]
appdb> db.users.find({ age: 26 })
[
  {
    _id: ObjectId("6112809d7ec144c25156c4dd"),
    name: 'Kyle',
    age: 26,
    hobbies: [ 'Weight Lifting', 'Bowling' ],
    address: { street: '123 Main St', city: 'New York City' }
  }
]

```

```

appdb> db.users.find({ name: "Kyle"}, { name: 1, age: 1 })
[
  { _id: ObjectId("6112809d7ec144c25156c4dd"), name: 'Kyle', age: 26 }
]
appdb> db.users.find({ name: "Kyle"}, { name: 1, age: 1, _id: 0 })
[ { name: 'Kyle', age: 26 } ]
appdb> db.users.find({ name: "Kyle"}, { age: 0 })
[
  {
    _id: ObjectId("6112809d7ec144c25156c4dd"),
    name: 'Kyle',
    hobbies: [ 'Weight Lifting', 'Bowling' ],
    address: { street: '123 Main St', city: 'New York City' }
  }
]

```

```
appdb> db.users.find({ name: { $eq: "Sally" } })
[
  {
    _id: ObjectId("611280447ec144c25156c4da"),
    name: 'Sally',
    age: 19,
    address: { street: '987 North St' },
    hobbies: [ 'Running' ]
  }
]
```

```
appdb> db.users.find({ name: { $neq: "Sally" } })
MongoServerError: unknown operator: $neq
appdb> db.users.find({ name: { $ne: "Sally" } })
[
  { _id: ObjectId("61127fc57ec144c25156c4d9"), name: 'John' },
  { _id: ObjectId("611280837ec144c25156c4db"), name: 'Jill' },
  { _id: ObjectId("611280837ec144c25156c4dc"), name: 'Mike' },
  {
    _id: ObjectId("6112809d7ec144c25156c4dd"),
    name: 'Kyle',
    age: 26,
    hobbies: [ 'Weight Lifting', 'Bowling' ],
    address: { street: '123 Main St', city: 'New York City' }
  },
  {
    _id: ObjectId("6112809d7ec144c25156c4de"),
    name: 'Billy',
    age: 41,
    hobbies: [ 'Swimming', 'Bowling' ],
    address: { street: '442 South St', city: 'New York City' }
  }
]
```



```
appdb> db.users.find({ age: { $gt: 13 } })
[
  {
    _id: ObjectId("611280447ec144c25156c4da"),
    name: 'Sally',
    age: 19,
    address: { street: '987 North St' },
    hobbies: [ 'Running' ]
  },
  {
    _id: ObjectId("6112809d7ec144c25156c4dd"),
    name: 'Kyle',
    age: 26,
    hobbies: [ 'Weight Lifting', 'Bowling' ],
    address: { street: '123 Main St', city: 'New York City' }
  },
  {
    _id: ObjectId("6112809d7ec144c25156c4de"),
    name: 'Billy',
    age: 41,
    hobbies: [ 'Swimming', 'Bowling' ],
    address: { street: '442 South St', city: 'New York City' }
  }
]
```

```
appdb> db.users.find({ age: { $lte: 19 } })
[
  {
    _id: ObjectId("611280447ec144c25156c4da"),
    name: 'Sally',
    age: 19,
    address: { street: '987 North St' },
    hobbies: [ 'Running' ]
  }
]
```



```
appdb> db.users.find({ name: { $in: ["Kyle", "Sally"] }})
[
  {
    _id: ObjectId("611280447ec144c25156c4da"),
    name: 'Sally',
    age: 19,
    address: { street: '987 North St' },
    hobbies: [ 'Running' ]
  },
  {
    _id: ObjectId("6112809d7ec144c25156c4dd"),
    name: 'Kyle',
    age: 26,
    hobbies: [ 'Weight Lifting', 'Bowling' ],
    address: { street: '123 Main St', city: 'New York City' }
  }
]
```

```
appdb> db.users.find({ name: { $nin: ["Kyle", "Sally"] }})
[
  { _id: ObjectId("61127fc57ec144c25156c4d9"), name: 'John' },
  { _id: ObjectId("611280837ec144c25156c4db"), name: 'Jill' },
  { _id: ObjectId("611280837ec144c25156c4dc"), name: 'Mike' },
  {
    _id: ObjectId("6112809d7ec144c25156c4de"),
    name: 'Billy',
    age: 41,
    hobbies: [ 'Swimming', 'Bowling' ],
    address: { street: '442 South St', city: 'New York City' }
  }
]
```

```

appdb> db.users.find({ age: { $exists: true }})
[
  {
    _id: ObjectId("611280447ec144c25156c4da"),
    name: 'Sally',
    age: 19,
    address: { street: '987 North St' },
    hobbies: [ 'Running' ]
  },
  {
    _id: ObjectId("6112809d7ec144c25156c4dd"),
    name: 'Kyle',
    age: 26,
    hobbies: [ 'Weight Lifting', 'Bowling' ],
    address: { street: '123 Main St', city: 'New York City' }
  },
  {
    _id: ObjectId("6112809d7ec144c25156c4de"),
    name: 'Billy',
    age: 41,
    hobbies: [ 'Swimming', 'Bowling' ],
    address: { street: '442 South St', city: 'New York City' }
  }
]
appdb> db.users.find({ age: { $exists: true }})

```

```

appdb> db.users.find({ age: { $exists: false }})
[
  { _id: ObjectId("61127fc57ec144c25156c4d9"), name: 'John' },
  { _id: ObjectId("611280837ec144c25156c4db"), name: 'Jill' },
  { _id: ObjectId("611280837ec144c25156c4dc"), name: 'Mike' }
]

```

```
appdb> db.users.find({ age: { $gte: 20, $lte: 40 } })
[
  {
    _id: ObjectId("6112809d7ec144c25156c4dd"),
    name: 'Kyle',
    age: 26,
    hobbies: [ 'Weight Lifting', 'Bowling' ],
    address: { street: '123 Main St', city: 'New York City' }
  }
]
```

```
appdb> db.users.find({ age: { $gte: 20, $lte: 40 }, name: "Kyle" })
[
  {
    _id: ObjectId("6112809d7ec144c25156c4dd"),
    name: 'Kyle',
    age: 26,
    hobbies: [ 'Weight Lifting', 'Bowling' ],
    address: { street: '123 Main St', city: 'New York City' }
  }
]
```

```
appdb> db.users.find({ $and: [{ age: 26 }, { name: "Kyle" }] })
[
  {
    _id: ObjectId("6112809d7ec144c25156c4dd"),
    name: 'Kyle',
    age: 26,
    hobbies: [ 'Weight Lifting', 'Bowling' ],
    address: { street: '123 Main St', city: 'New York City' }
  }
]
```

```
appdb> db.users.find({ $or: [{ age: { $lte: 20 } }, { name: "Kyle" }] })
[
  {
    _id: ObjectId("611280447ec144c25156c4da"),
    name: 'Sally',
    age: 19,
    address: { street: '987 North St' },
    hobbies: [ 'Running' ]
  },
  {
    _id: ObjectId("6112809d7ec144c25156c4dd"),
    name: 'Kyle',
    age: 26,
    hobbies: [ 'Weight Lifting', 'Bowling' ],
    address: { street: '123 Main St', city: 'New York City' }
  }
]
```

```
appdb> db.users.find({ age: { $not: { $lte: 20 } } })
[
  { _id: ObjectId("61127fc57ec144c25156c4d9"), name: 'John' },
  { _id: ObjectId("611280837ec144c25156c4db"), name: 'Jill' },
  { _id: ObjectId("611280837ec144c25156c4dc"), name: 'Mike' },
  {
    _id: ObjectId("6112809d7ec144c25156c4dd"),
    name: 'Kyle',
    age: 26,
    hobbies: [ 'Weight Lifting', 'Bowling' ],
    address: { street: '123 Main St', city: 'New York City' }
  },
  {
    _id: ObjectId("6112809d7ec144c25156c4de"),
    name: 'Billy',
    age: 41,
    hobbies: [ 'Swimming', 'Bowling' ],
    address: { street: '442 South St', city: 'New York City' }
  },
  {
    _id: ObjectId("611283c07ec144c25156c4df"),
    age: null,
    name: 'Bill'
  }
]
```



```

appdb> db.users.find({ age: {$gt: 20 }})
[
  {
    _id: ObjectId("6112809d7ec144c25156c4dd"),
    name: 'Kyle',
    age: 26,
    hobbies: [ 'Weight Lifting', 'Bowling' ],
    address: { street: '123 Main St', city: 'New York City' }
  },
  {
    _id: ObjectId("6112809d7ec144c25156c4de"),
    name: 'Billy',
    age: 41,
    hobbies: [ 'Swimming', 'Bowling' ],
    address: { street: '442 South St', city: 'New York City' }
  }
]

```

```

appdb> db.users.insertMany([ { name: "Tom", balance: 100, debt: 200 }, { name: "Kristin", balance: 20, debt: 0 } ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId("6112858b7ec144c25156c4e0"),
    '1': ObjectId("6112858b7ec144c25156c4e1")
  }
}

```

```

appdb> db.users.find({ $expr: { $gt: [ "$debt", "$balance" ] } })
[
  {
    _id: ObjectId("6112858b7ec144c25156c4e0"),
    name: 'Tom',
    balance: 100,
    debt: 200
  }
]

```

```
appdb> db.users.find({ "address.street": "123 Main St" })
[
  {
    _id: ObjectId("6112809d7ec144c25156c4dd"),
    name: 'Kyle',
    age: 26,
    hobbies: [ 'Weight Lifting', 'Bowling' ],
    address: { street: '123 Main St', city: 'New York City' }
  }
]
```

```
appdb> db.users.findOne({ age: { $lte: 40 } })
{
  _id: ObjectId("611280447ec144c25156c4da"),
  name: 'Sally',
  age: 19,
  address: { street: '987 North St' },
  hobbies: [ 'Running' ]
}
```

```
appdb> db.users.countDocuments({ age: { $lte: 40 } })
2
```

Updating of record:

```
appdb> db.users.updateOne({age: 26}, { $set: { age: 27 } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
appdb> db.users.findOne({ age: 27 })
{
  _id: ObjectId("6112809d7ec144c25156c4dd"),
  name: 'Kyle',
  age: 27,
  hobbies: [ 'Weight Lifting', 'Bowling' ],
  address: { street: '123 Main St', city: 'New York City' }
}
```

```
appdb> db.users.updateOne({ _id: ObjectId("6112809d7ec144c25156c4dd") }, { $set: { name: "New Name" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
appdb> db.users.findOne({ _id: ObjectId("6112809d7ec144c25156c4dd") })
{
  _id: ObjectId("6112809d7ec144c25156c4dd"),
  name: 'New Name',
  age: 27,
  hobbies: [ 'Weight Lifting', 'Bowling' ],
  address: { street: '123 Main St', city: 'New York City' }
}
```

```

appdb> db.users.updateOne({ _id: ObjectId("6112809d7ec144c25156c4dd") }, { $inc: { age: 3 } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
appdb> db.users.findOne({ _id: ObjectId("6112809d7ec144c25156c4dd") })
{
  _id: ObjectId("6112809d7ec144c25156c4dd"),
  name: 'New Name',
  age: 30,
  hobbies: [ 'Weight Lifting', 'Bowling' ],
  address: { street: '123 Main St', city: 'New York City' }
}

```

```

appdb> db.users.updateOne({ _id: ObjectId("6112809d7ec144c25156c4dd") }, { $rename: { name: "firstName" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
appdb> db.users.findOne({ _id: ObjectId("6112809d7ec144c25156c4dd") })
{
  _id: ObjectId("6112809d7ec144c25156c4dd"),
  age: 30,
  hobbies: [ 'Weight Lifting', 'Bowling' ],
  address: { street: '123 Main St', city: 'New York City' },
  firstName: 'New Name'
}

```

Unset (or remove the property age)

```

appdb> db.users.updateOne({ _id: ObjectId("6112809d7ec144c25156c4dd") }, { $unset: { age: "" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
appdb> db.users.findOne({ _id: ObjectId("6112809d7ec144c25156c4dd") })
{
  _id: ObjectId("6112809d7ec144c25156c4dd"),
  hobbies: [ 'Weight Lifting', 'Bowling' ],
  address: { street: '123 Main St', city: 'New York City' },
  firstName: 'New Name'
}

```

Push (Add "Swimming" to hobbies)

```

{
  _id: ObjectId("6112809d7ec144c25156c4dd"),
  hobbies: [ 'Weight Lifting', 'Bowling' ],
  address: { street: '123 Main St', city: 'New York City' },
  firstName: 'New Name'
}
appdb> db.users.updateOne({ _id: ObjectId("6112809d7ec144c25156c4dd") }, { $push: { hobbies: "Swimming" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
appdb> db.users.findOne({ _id: ObjectId("6112809d7ec144c25156c4dd") })
{
  _id: ObjectId("6112809d7ec144c25156c4dd"),
  hobbies: [ 'Weight Lifting', 'Bowling', 'Swimming' ],
  address: { street: '123 Main St', city: 'New York City' },
  firstName: 'New Name'
}

```

Pull (Remove “Bowling” from hobbies)

```

{
  _id: ObjectId("6112809d7ec144c25156c4dd"),
  hobbies: [ 'Weight Lifting', 'Bowling' ],
  address: { street: '123 Main St', city: 'New York City' },
  firstName: 'New Name'
}
appdb> db.users.updateOne({ _id: ObjectId("6112809d7ec144c25156c4dd") }, { $push: { hobbies: "Swimming" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
appdb> db.users.findOne({ _id: ObjectId("6112809d7ec144c25156c4dd") })
{
  _id: ObjectId("6112809d7ec144c25156c4dd"),
  hobbies: [ 'Weight Lifting', 'Bowling', 'Swimming' ],
  address: { street: '123 Main St', city: 'New York City' },
  firstName: 'New Name'
}

```

Pull i.e. Remove bowling:

```

appdb> db.users.updateOne({ _id: ObjectId("6112809d7ec144c25156c4dd") }, { $pull: { hobbies: "Bowling" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
appdb> db.users.findOne({ _id: ObjectId("6112809d7ec144c25156c4dd") })
{
  _id: ObjectId("6112809d7ec144c25156c4dd"),
  hobbies: [ 'Weight Lifting', 'Swimming' ],
  address: { street: '123 Main St', city: 'New York City' },
  firstName: 'New Name'
}

```


Remove Address:

```
appdb> db.users.updateMany({ address: { $exists: true } }, { $unset: { address: "" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
appdb> db.users.find()
```

Replace:

```
appdb> db.users.replaceOne({ age: 30 }, { name: "John" })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
appdb> db.users.find({ name: "John" })
[ { _id: ObjectId("61127fc57ec144c25156c4d9"), name: 'John' } ]
```

```
appdb> db.users.find()
[
  { _id: ObjectId("61127fc57ec144c25156c4d9"), name: 'John' },
  {
    _id: ObjectId("611280447ec144c25156c4da"),
    name: 'Sally',
    age: 19,
    hobbies: [ 'Running' ]
  },
  { _id: ObjectId("611280837ec144c25156c4db"), name: 'Jill' },
  { _id: ObjectId("611280837ec144c25156c4dc"), name: 'Mike' },
  {
    _id: ObjectId("6112809d7ec144c25156c4dd"),
    hobbies: [ 'Weight Lifting', 'Swimming' ],
    firstName: 'New Name'
  },
  {
    _id: ObjectId("6112809d7ec144c25156c4de"),
    name: 'Billy',
    age: 41,
    hobbies: [ 'Swimming', 'Bowling' ]
  },
  {
    _id: ObjectId("611283c07ec144c25156c4df"),
    age: null,
    name: 'Bill'
  },
]
```

Delete

```
appdb> db.users.replaceOne({ age: 30 }, { name: "John" })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
appdb> db.users.find({ name: "John" })
[ { _id: ObjectId("61127fc57ec144c25156c4d9"), name: 'John' } ]
```

```
appdb> db.users.deleteMany({ age: { $exists: false } })
{ acknowledged: true, deletedCount: 5 }
appdb> db.users.find()
[
  {
    _id: ObjectId("611280447ec144c25156c4da"),
    name: 'Sally',
    age: 19,
    hobbies: [ 'Running' ]
  },
  {
    _id: ObjectId("6112809d7ec144c25156c4de"),
    name: 'Billy',
    age: 41,
    hobbies: [ 'Swimming', 'Bowling' ]
  },
  {
    _id: ObjectId("611283c07ec144c25156c4df"),
    age: null,
    name: 'Bill'
  }
]
```

